Taeyoung Son

Name: Taeyoung Son (손태영)
Email: ty.son@postech.ac.kr
Phone: +8210-5755-6746

Personal Website
LinkedIn
Google Scholar
GitHub

I am a machine learning engineer who specializes in bridging the gap between state-of-the-art research and scalable, production-ready products. My work spans from recommendation systems and generative AI to computer vision, combining strong research experience with hands-on engineering expertise.

Publications

WEDGE: Web-Image Assisted Domain Generalization for Semantic Segmentation

*Namyup Kim, **Taeyoung Son**, Jaehyun Pahk, Cuiling Lan, Wenjun Zeng, Suha Kwak*

*IEEE International Conference on Robotics and Automation (**ICRA**), 2023*

arxiv: https://arxiv.org/abs/2109.14196

FIFO: Learning Fog-Invariant Features for Foggy Scene Segmentation

*Sohyun Lee, **Taeyoung Son**, Suha Kwak*

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022 (**Oral**, Best Paper Finalist)*

arxiv: https://arxiv.org/abs/2204.01587

URIE: Universal Image Enhancement for Visual Recognition in the Wild

***Taeyoung Son**, Juwon Kang, Namyup Kim, Sunghyun Cho, Suha Kwak*

*European Conference on Computer Vision (**ECCV**), 2020*

arxiv: https://arxiv.org/abs/2007.08979

Patent

Method and device for learning fog-invariant feature

- *Sohyun Lee, Suha Kwak, **Taeyoung Son***
- US Patent number: *US20230419654A1*

Skills

Python, Pytorch, SQL, Generative AI, LLM, Apache Spark, Apache Airflow, AWS

Education

Mar. 2020 – Mar. 2022

M.S., POSTECH (Pohang University of Science and Technology)

- Department of Computer Science and Engineering (CSE)
- Computer Vision Laboratory
- Thesis: *Universal Image Enhancement for Robust Visual Recognition*
- Advisor: Prof. Suha Kwak

Mar. 2015 – Feb. 2020

B.S., POSTECH (Pohang University of Science and Technology)

- Department of Computer Science and Engineering (CSE)
- Thesis: *Visual recognition under extreme condition*
- Advisor: Prof. Suha Kwak

Careers

Machine Learning Engineer at *RIDI*

Oct. 2023 – Present

Developed a recommendation system for the 'For You' section of the Manta

Apr.2025 - Present

- Engineered a data pipeline using Airflow and Kafka to periodically or in real-time stream source data into AWS S3.
- Built and managed a feature store, organizing data by source and feature units, to consistently generate and manage datasets for training the recommendation system.
- Collaborated with project managers and data analysts to create features based on various project initiatives. Continuously trained recommendation models and conducted online A/B tests to roll out winning versions.
- Implemented models such as VAE-CF, BERT4REC, and SASREC for recommendation model training and established a pipeline for their training on AWS SageMaker.
- Set up an MLflow tracking server integrated with AWS SageMaker to monitor the training process and manage the lifecycle of experiments.
- Served personalized recommendations to hundreds of thousands of monthly active users by processing gigabytes of interaction data on a regular basis.

- **Achievement:** Improved the Click-Through Rate (CTR) of the Manta 'For You' tab recommendations by 13 percentage points.

Developed the 'Trending Now' recommendation feature for the RIDIBOOKS

Jan. 2025 - Apr. 2025

- Implemented a batch process to dump user search data to S3 and aggregate the most searched works within a defined period using Spark.
- Automated the collection and deployment of the aggregated data to be displayed on the search page of the RIDIBOOKS service at regular intervals.
- **Achievement:** Increased page conversion rate by 1.79%p and paid conversion rate by 1.00%p without negatively impacting Average Revenue Per User (ARPU).

Developed a chatbot to enhance internal knowledge accessibility

Oct.2024 - Feb. 2025

- Developed 'GONYAGI', a real-time translation chatbot integrated with Slack, to facilitate seamless communication among multilingual teams (Japanese, English, Korean).
- Built a Retrieval Augmented Generation (RAG) chatbot by embedding internal Notion documents into an Azure AI Search vector database, enabling it to answer employee queries by retrieving relevant information.
- **Achievement:** Significantly reduced the time spent on repetitive internal inquiries by automating responses through the AI chatbot.

Developed an image generation algorithm for the Prodifi

Oct.2023 - Oct. 2024

- Implemented a ControlNet training and evaluation pipeline to generate images based on various conditions such as pose, depth, scribble, and masks.
- Built a LoRA training and evaluation pipeline to consistently generate images of characters from the company's proprietary IP.
- Established a ViTPose model training and evaluation pipeline to construct a pose dataset from in-house content.
- Developed an internal image generation testing tool using ComfyUI and Stable Diffusion WebUI.
- **Contribution:** Enabled the internal content team to efficiently control the pose and composition of generated images using pose-controlnet. Contributed to the effective filling of speech bubbles and background effects during localization tasks in the Prodifi tool using inpaint-controlnet.

AI Research Scientist at NALBI

Apr. 2022 – Oct. 2023

Developed a real-time 3D human body reconstruction algorithm for the A.PLA

- Implemented the Part Attention Regressor for 3D Human Body Estimation (PARE) model and its training/evaluation pipeline to extract human motion data from videos in the Skinned Multi-Person Linear Model (SMPL) format.
- Implemented an automated SMPL annotation pipeline using ROKOKO motion capture equipment for dataset construction.
- Optimized the model for real-time inference on mobile and CPU-only environments using Post-Training Quantization techniques, achieving over 30 FPS across various devices.

Research Scientist Internship at HyperConnect

Jun. 2018 – Sep. 2018, Jun. 2019 – Aug. 2019

Trained a classification model to filter inappropriate profile pictures in the Azar

Jun. 2019 – Aug. 2019

- Developed an image classification model to automatically filter inappropriate user profile pictures from a large-scale, imbalanced dataset of approximately 4.6M of images.
- Improved the existing Inception-ResNet V2 model by replacing it with EfficientNet-V5, achieving an accuracy of 96.51%.
- Researched and applied various self-supervised learning techniques, such as Data Distillation and Label Refinery, to address data imbalance and noise issues.

Developed a model to generate Look-Up Tables (LUTs) for on-device camera beautification filters

Jun. 2018 – Sep. 2018

- Implemented a web crawler to collect highly-rated images from Flickr.
- Developed a pixel-wise LUT generator for image enhancement and color correction.

Personal Projects

A collection of side projects based on personal interests.

Rotary Position Embedding in Triton

- As part of a kernel optimization study, I implemented an optimized GPU kernel for Rotary Position Embedding (RoPE) using the Triton language.
- Validated the performance improvements of the custom-built kernel through benchmarking against existing implementations.
- *Github: https://github.com/taeyoungson/triton_rope*

Flight Schedule Manager with AI Weather Assistant

To efficiently manage my girlfriend's flight attendant schedule, I automated the following process:

- Used iPhone's "Shortcuts" feature to perform OCR on schedule photos and send the data to a FastAPI server deployed on an EC2 instance.
- Extracted departure/arrival locations and times from the parsed data and registered them on Google Calendar.
- Queried for flight delays 20 minutes before each scheduled arrival to provide the final estimated arrival time.
- For international destinations, an LLM provides a summary of the local weather and suggested attire 1-2 days before departure.
- *Github: https://github.com/taeyoungson/flight-schedule-parser*

AI-Powered Automated Stock Trader

- Utilized an LLM to parse and analyze financial statements from DART (Data Analysis, Retrieval and Transfer System) to select stocks for investment.
- Implemented automated trading strategies for the selected stocks, executing buy and sell orders.
- *Github: https://github.com/taeyoungson/trader*

Honors & Awards

- CVPR Best Paper Finalist, 2022 (Awarded to Top 0.4%)
- Winner, Qualcomm Innovation Fellowship 2022
- Seoul MaaS Hackathon Excellence Award 2019 (우수상)

Media

- https://news.mt.co.kr/mtview.php?no=2020091410243790785
- https://www.yeongnam.com/web/view.php?key=20220428010003756
- https://www.youtube.com/watch?v=LCV7OD3qUcQ